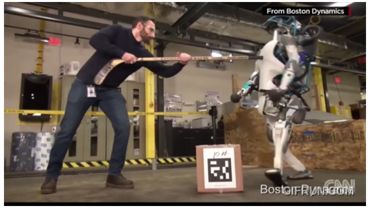

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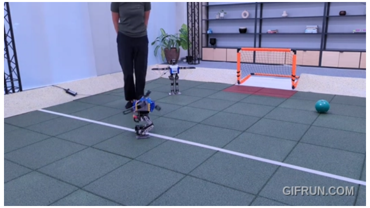

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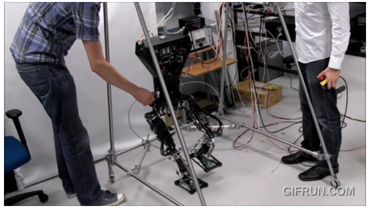

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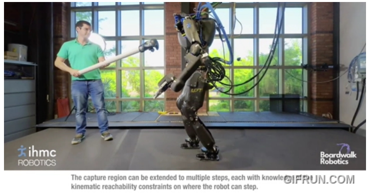

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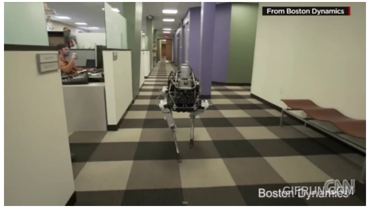


12

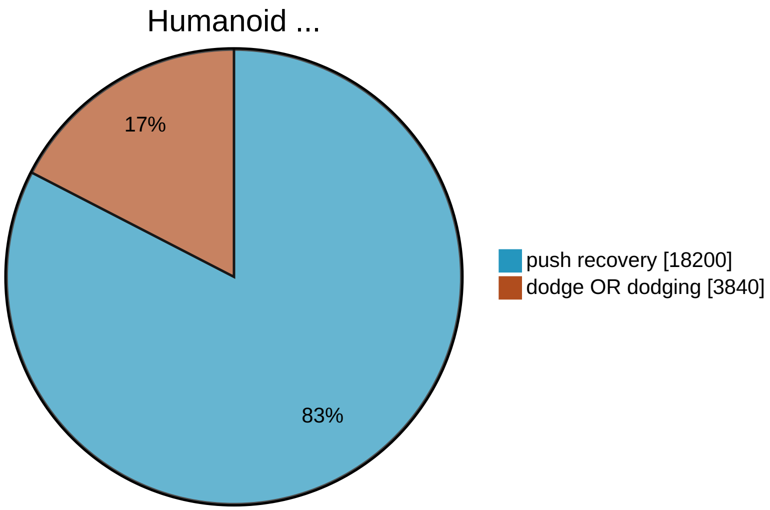
[1]CNN [2]OP3 Soccer [3]TU Delft [4]CNN [5]OSU [6]Unitree [7]OCP3 Soccer [8]CNN [9]Corridor Digital [10]IHMC Robotics [11]Agility Robotics [12]CNN

PROJECT NEO

Getting **He1**nz to dodge like the chosen one.

Dominic Zahn, 8/29/2025

Dodging is Underrepresented



Why Dodging Bullets is Intresting

No Killer-Robots ×

Effective Evasion Manauever ✅

Avoids Damage ✅

Investigating Stability Criterias ✅

Applyed Mathematical Optimization ✅

Exploring Platform Limitations ✅

> *"I don`t like to get hit, who likes it?"*
>
> – Wladimir Klitschko

Outline

Unitree H1 ↔ **Powerfull Platform** 🦾

Docker ↔ **Portable Environment** 📦

ROS 2 ↔ *Jazzy* 🧪

Toolbox 🧰

Optimization ↔ *NLOPT* 📈

Dynamics ↔ *RBDL* 🦿

Optimal Control ↔ *Bioptim* 🕹️

Vision ↔ *OpenCv* 🥽

[1]filmfreedonia.com

Table of Contents

Optimization Problem 📈

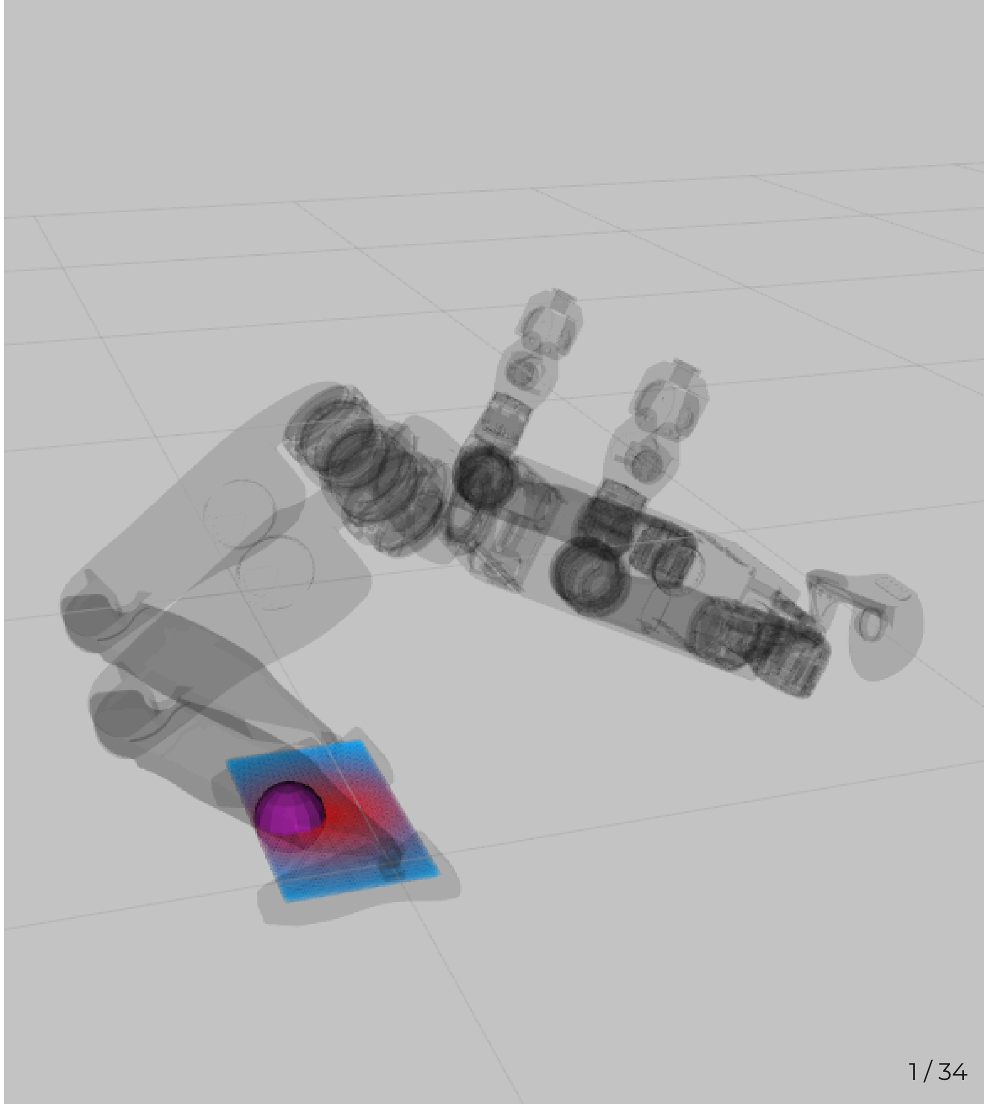
Problem:

- **avoid obstacles** on head level
- **feet** are static on the ground

=> $\mathrm{const.}$ Polygon of Support

- **statically stable**

=> $\mathrm{CoP} \in \mathrm{PoS}$



Acronyms

Center of Preassure = CoP
Polygon of Support = PoS

Optimization Problem 📈

Problem:

- **avoid obstacles** on head level
- **feet** are static on the ground

=> $\mathrm{const.}$ Polygon of Support

- **statically stable**

=> $\mathrm{CoP} \in \mathrm{PoS}$

Acronyms

Center of Preassure = CoP
Polygon of Support = PoS
Center of Mass = CoM

Mathematical Formulation (I)

$$p_{\text{head}}^{\text{world}} \neq p_{\text{obstacle}}^{\text{world}} \tag{1}$$

$$p_{\text{leftFoot}}^{\text{world}} \in \mathrm{GND} \tag{2}$$

$$\mathrm{AND} \quad p_{\text{rightFoot}}^{\text{world}} \in \mathrm{GND} \tag{3}$$

$$p_{\text{CoP}}^{\text{world}} \in \mathrm{PoS}^{\text{world}} \tag{4}$$

$$\xRightarrow{\text{static}} p_{\text{CoP}}^{\text{world}} := \begin{bmatrix} p_{\text{CoM}}^{\text{world}}.x \\ p_{\text{CoM}}^{\text{world}}.y \\ 0 \end{bmatrix} \tag{5}$$

Static Stability ⚖️

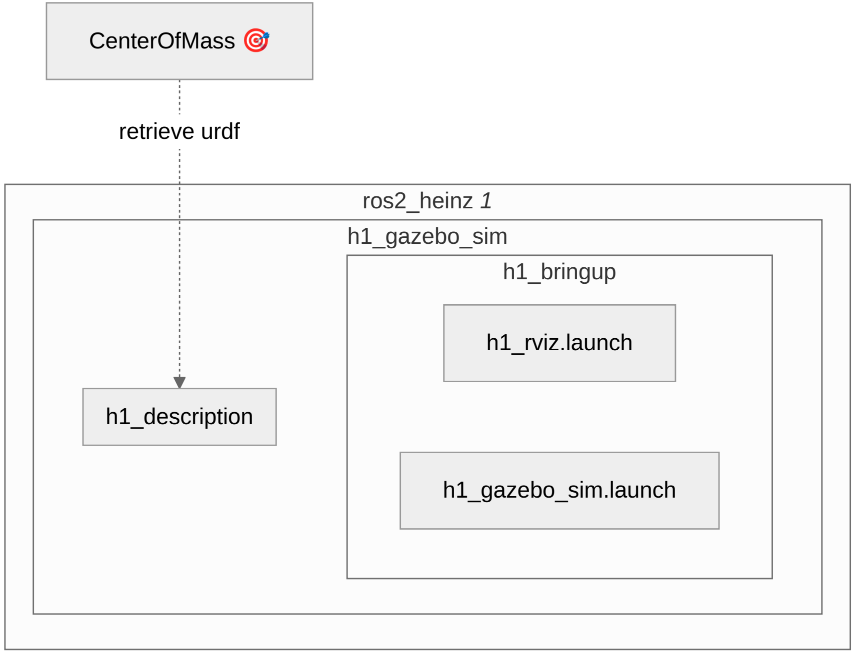
[1]Unitree

Center of Mass (CoM) 🎯

ros2_heinz

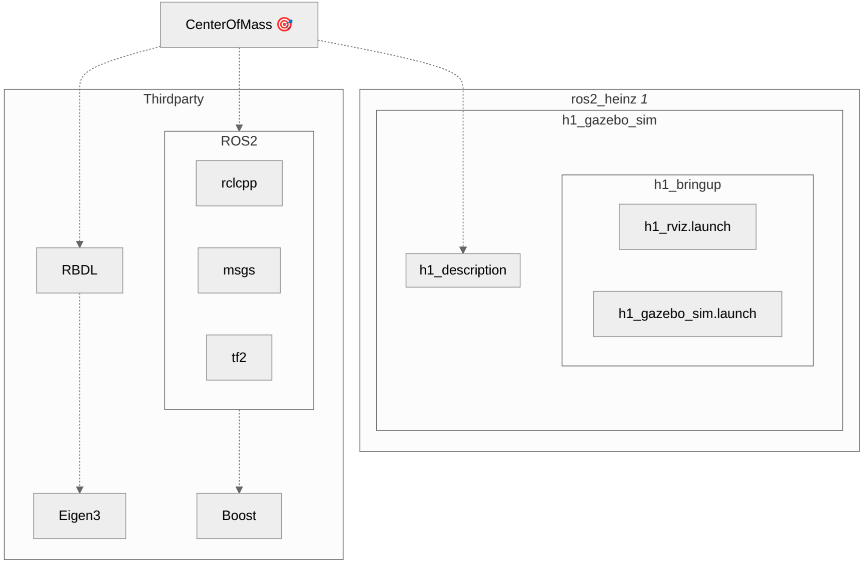
Simulate the H1 robot in ROS2 Jazzy and Gazebo Harmonic!

Python ☆ 7



[1] K-d4wg

Center of Mass (CoM) 🎯



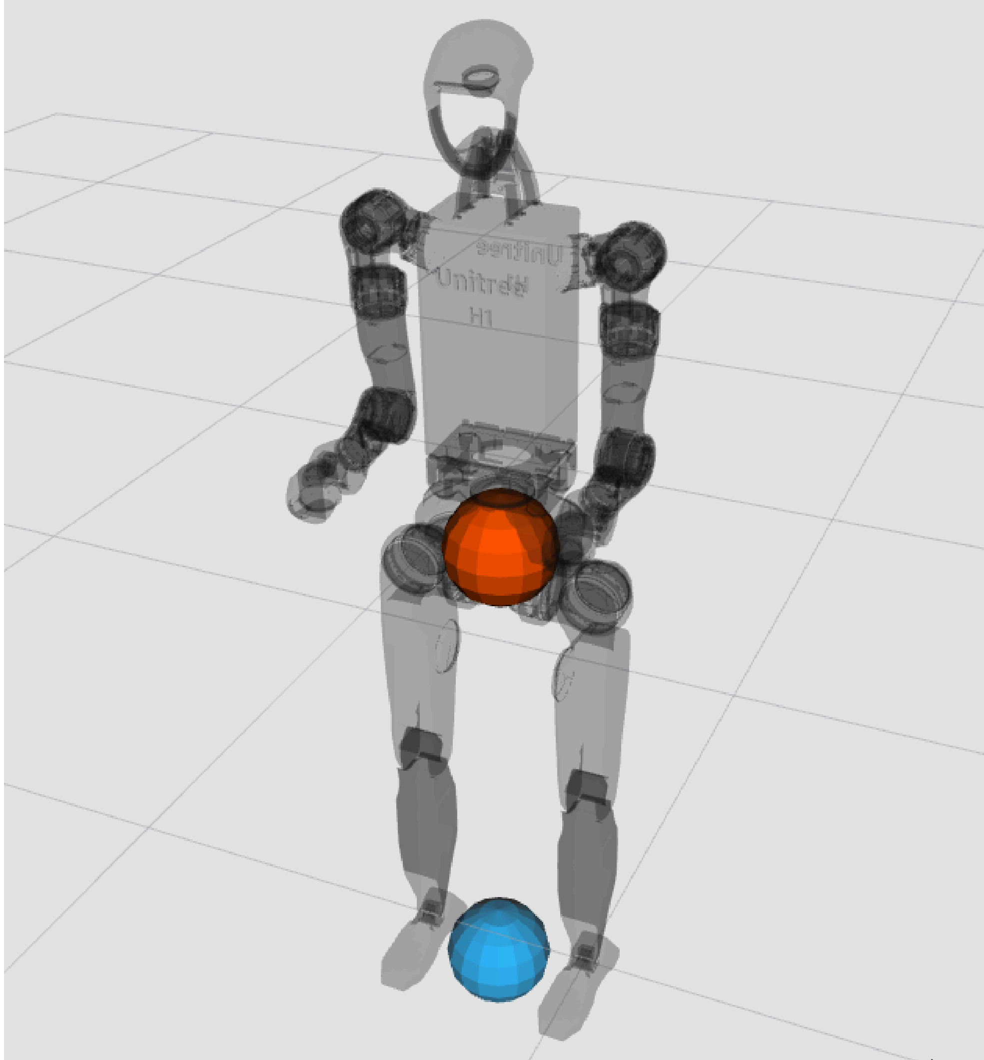
🔊 Publisher: `/CoM` $\xRightarrow{z=0}$ `/CoP`

Center of Mass
(CoM) 🎯

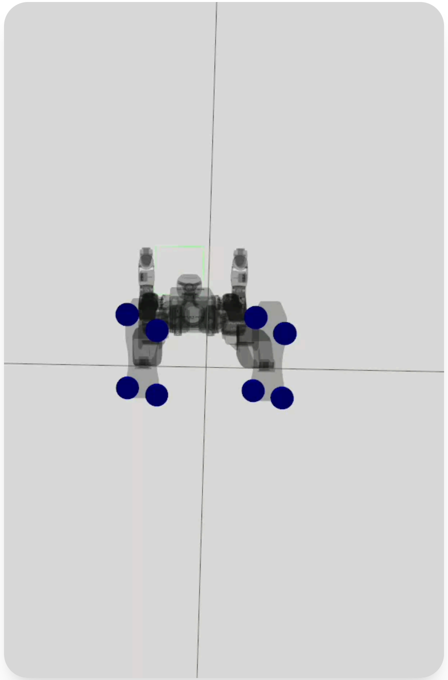
Legend

CoM 🟠

CoP 🔵

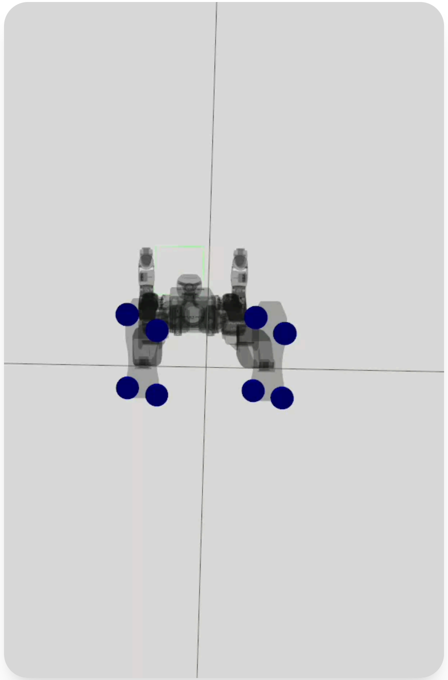


Polygon of Support (PoS) 👣



🔊Publisher: /PoS

Polygon of Support (PoS) 👣



🔊Publisher: $\texttt{/PoS} \xRightarrow{\text{Rviz}} \texttt{/vis\_PoS}$

Gazebo $\Longrightarrow$ Rviz

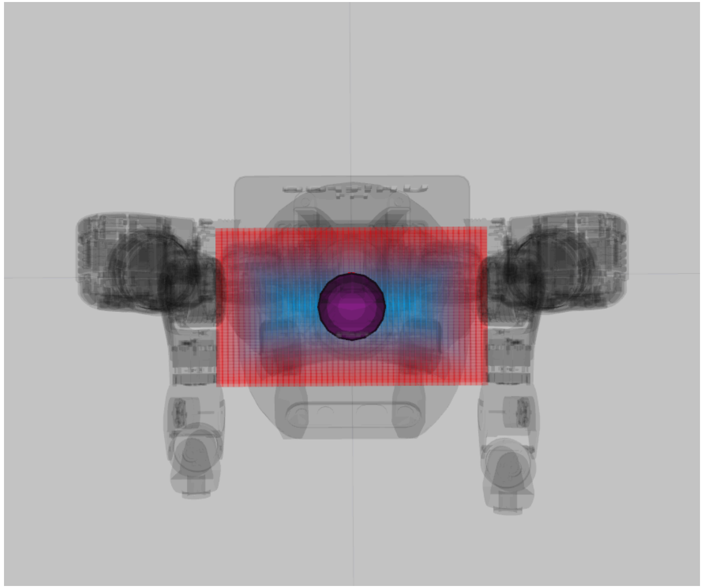
Flexibel Joint Control ✅
Full Visualization ✅
No Jiggeling ✅
Better Performance ✅

No Physics ×
No Fixed Global Frame ×
No Live Contacts ×

$\Longrightarrow$ Fixed $\mathbf{PoS}$ 🔒

Stability Criteria 📐

🔴: low values | 🔵: high values

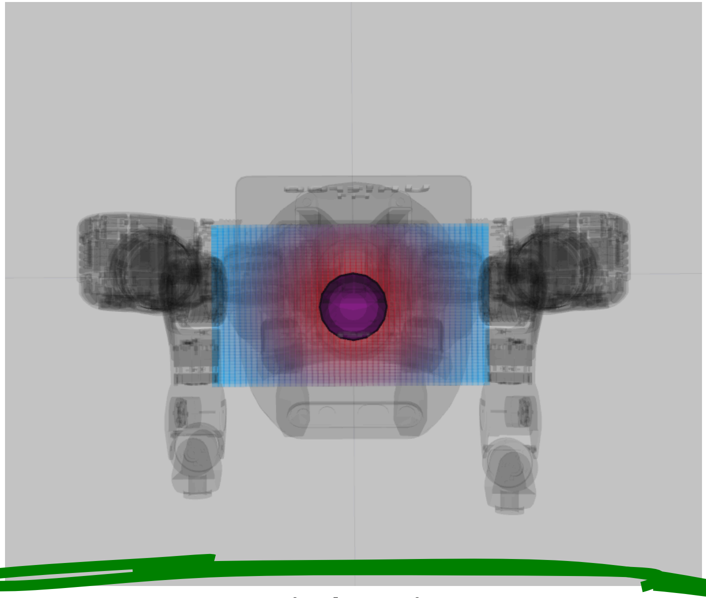


Min Edge Distance

🚫 No Continuity

✅ Uses PoS Borders

✅ Adjusts to Shape



Centroid Distance

✅ Continuous

🚫 Ignores PoS Borders

⚠️ Only Indirect Influence of Shape

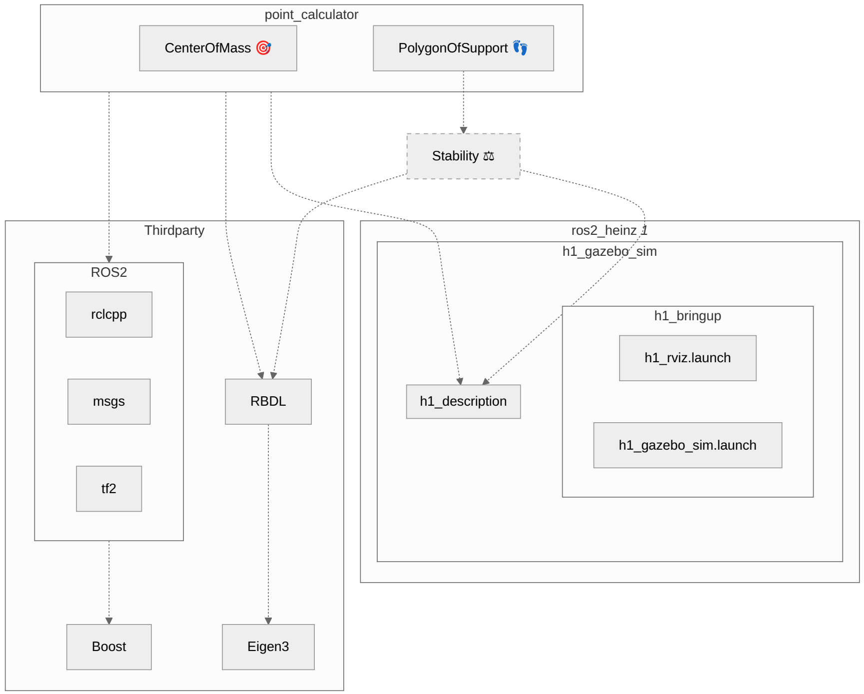
Stability ⚖️

📚Stability Library

- **Stability Criteria**
- **Helper Functions**

🔗Depends on

- **RBDL**
- **h1_description**



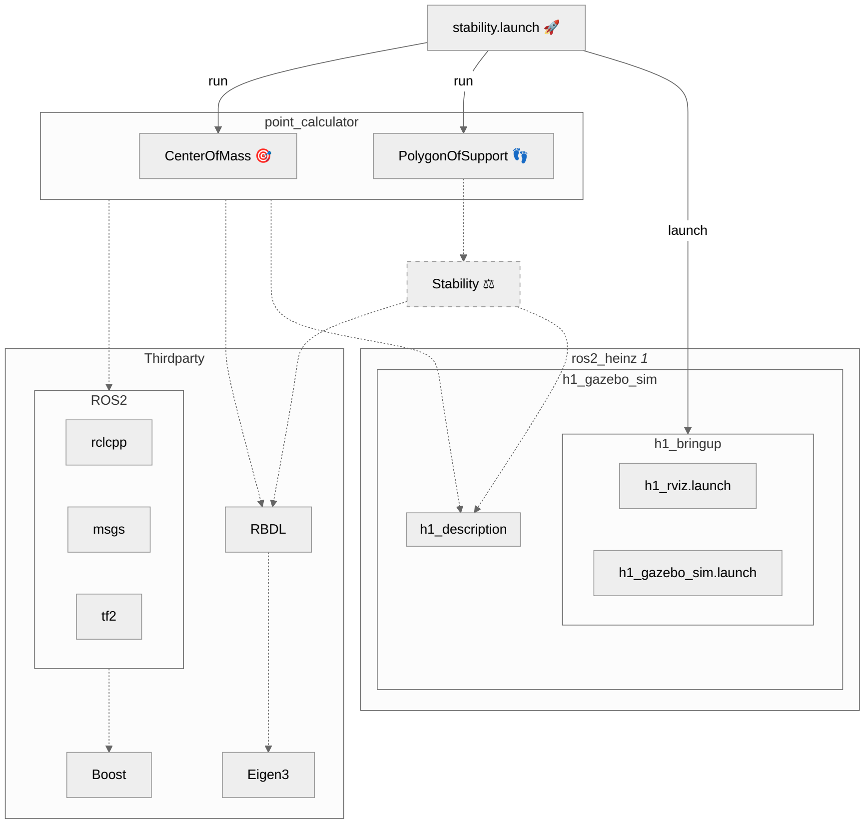
Launch Files

🚀 stability.launch

run **CenterOfMass** 🎯

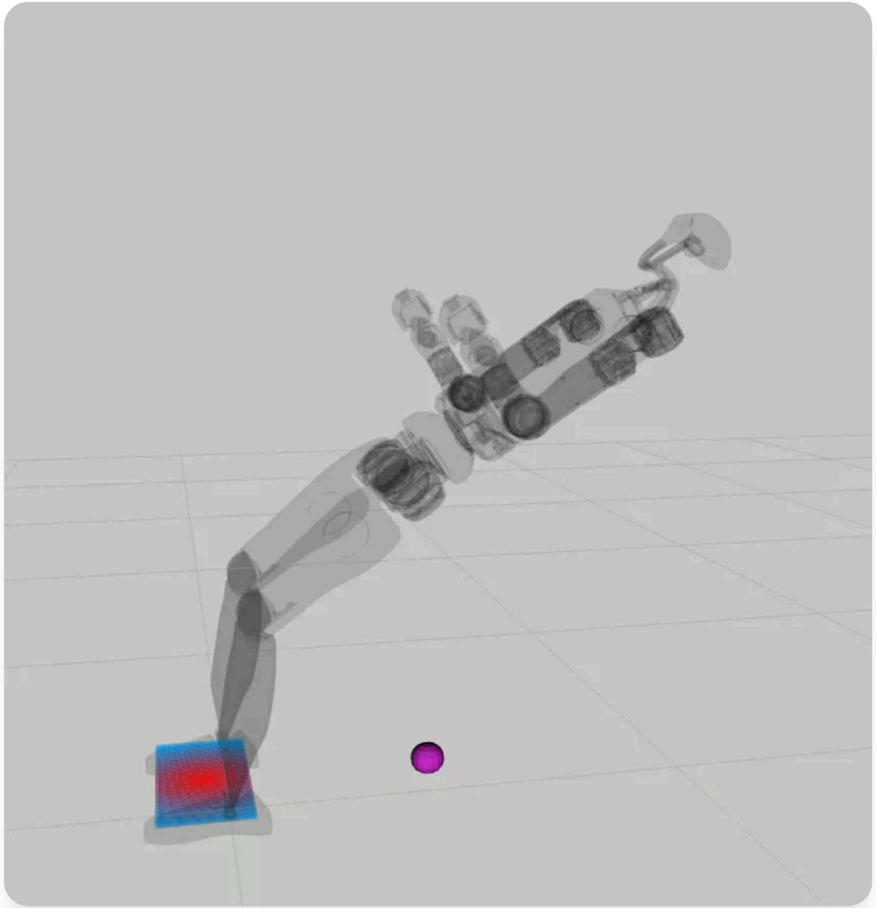
run **PolygonOfSupport** 👣

launch **rviz** *OR* **gazebo**🧮

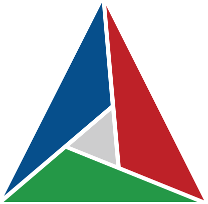


Actuation 🕹️

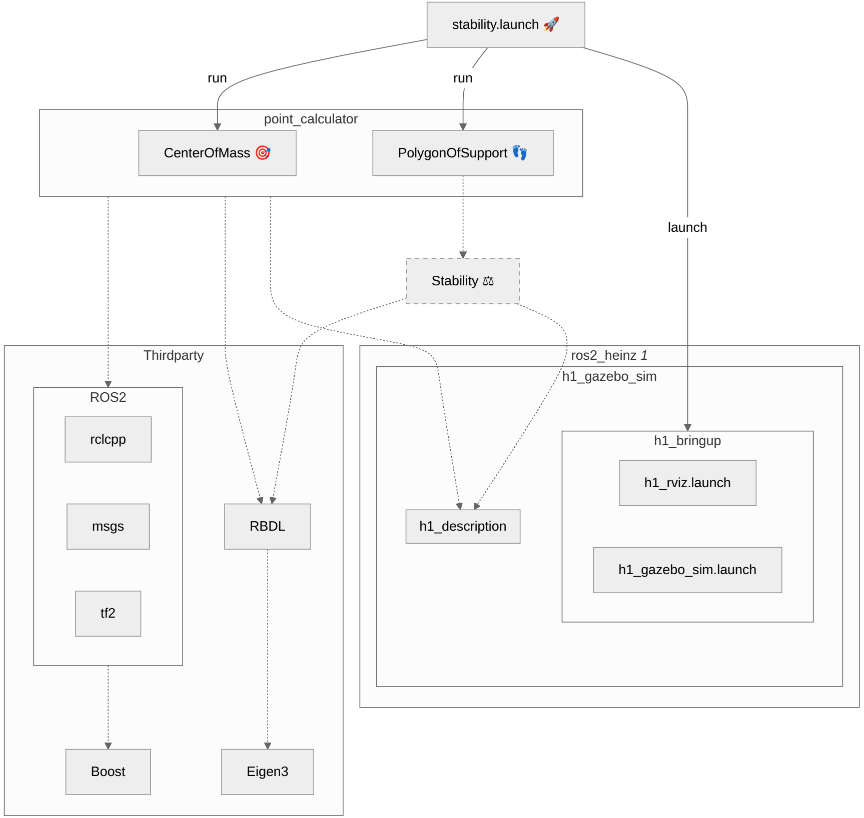
🔊Publishing to */joint_states*
- 🍎**Ignore Physics**
- ⏱️**Instant Movement**



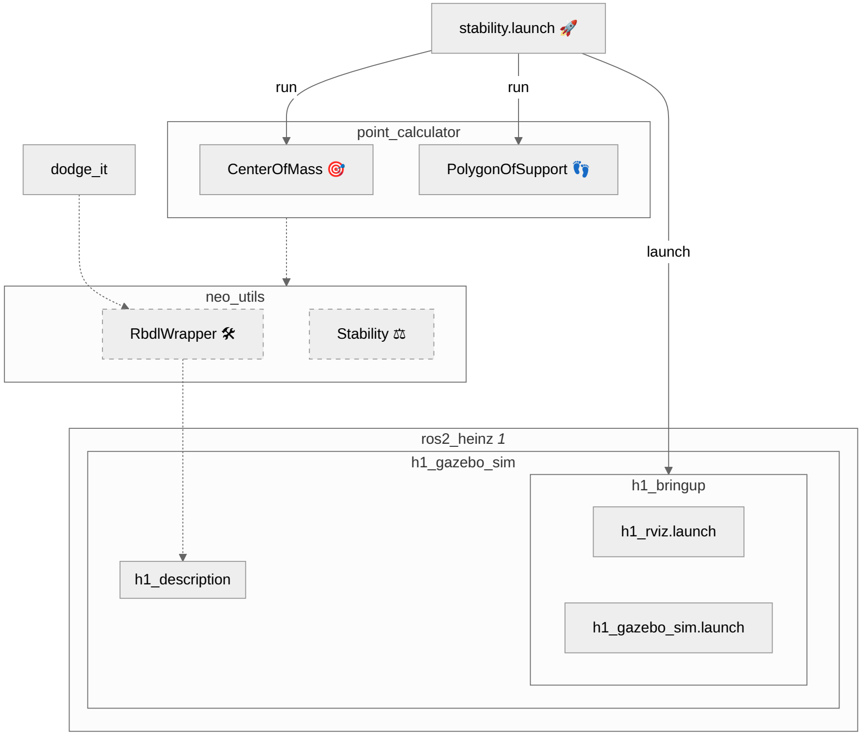
Dependencies 🔗

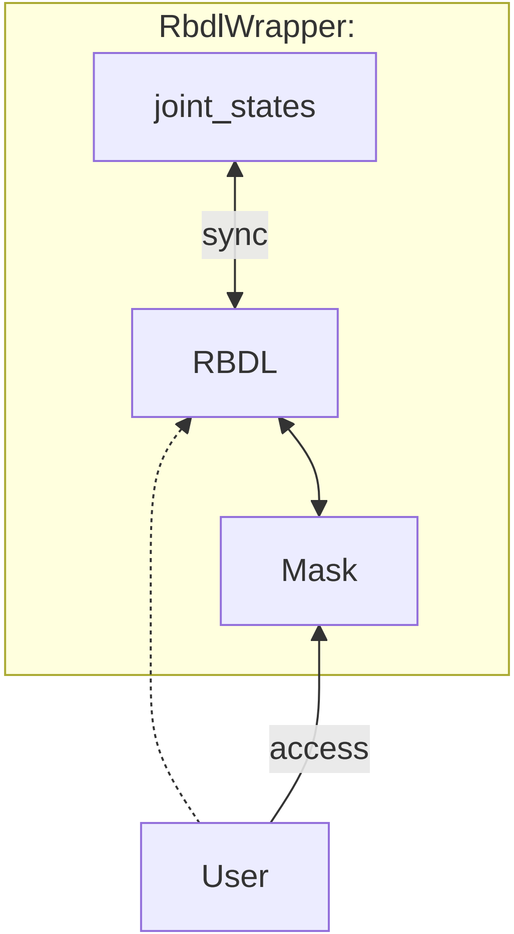


🚨 Dependency Chaos 🚨

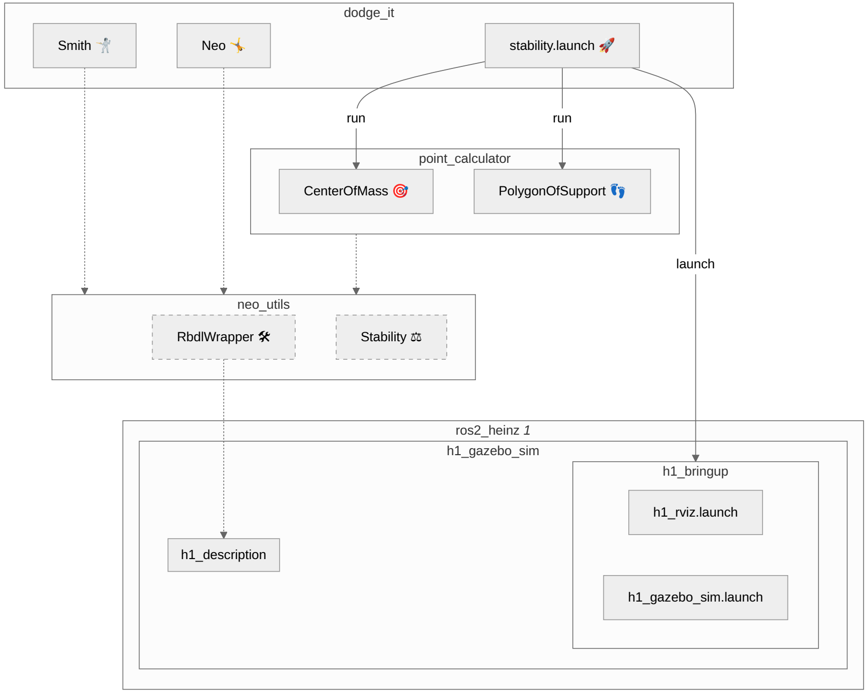


RbdlWrapper 🛠️





Framework Overview 🧰



[1] K-d4wg

Tech Stack Overview 📦

Running Optimization 🎲

Root $\neq$ World

Optimization **(I)**

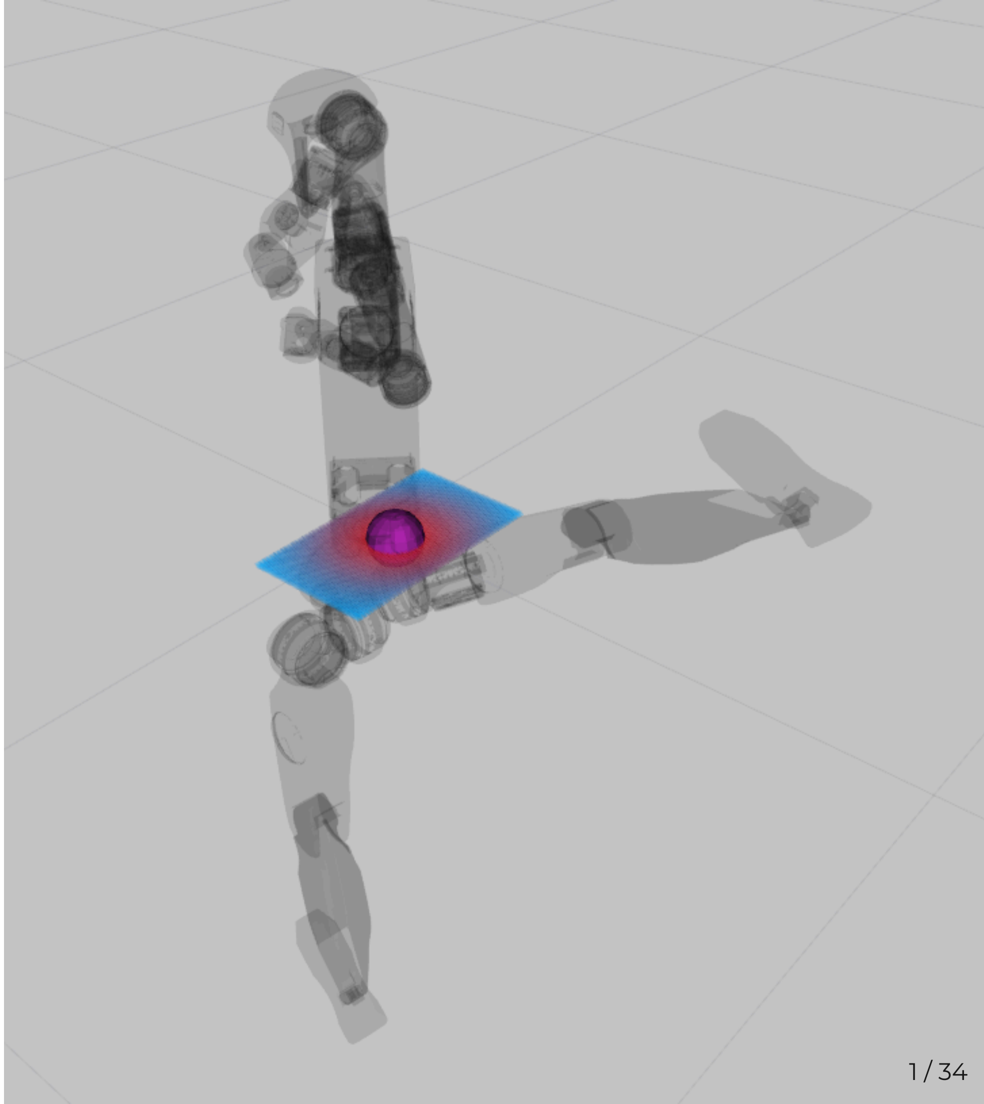
$$p^{\text{world}}_{\text{head}} \neq p^{\text{world}}_{\text{obstacle}} \tag{6}$$

$$p^{\text{world}}_{\text{leftFoot}} \in \text{GND} \tag{7}$$

$$\text{AND} \quad p^{\text{world}}_{\text{rightFoot}} \in \text{GND} \tag{8}$$

$$p^{\text{world}}_{\text{CoP}} \in \text{PoS}^{\text{world}} \tag{9}$$

$$\xRightarrow{\text{static}} p^{\text{world}}_{\text{CoP}} := \begin{bmatrix} p^{\text{world}}_{\text{CoM}}.x \\ p^{\text{world}}_{\text{CoM}}.y \\ 0 \end{bmatrix} \tag{10}$$



Root $\neq$ World

Optimization **(II)**

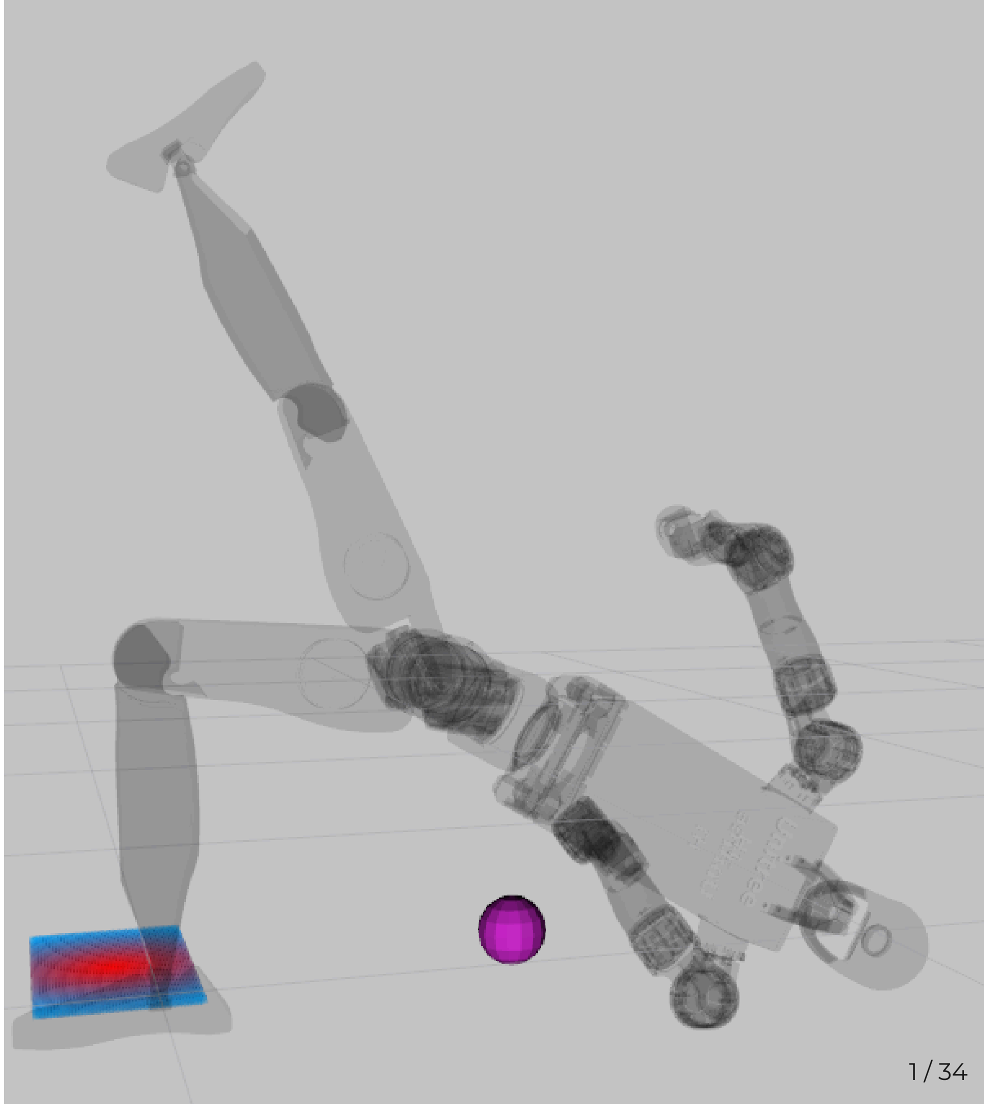
$$p_{\text{head}}^{\text{leftFoot}} \neq p_{\text{obstacle}}^{\text{leftFoot}} \tag{11}$$

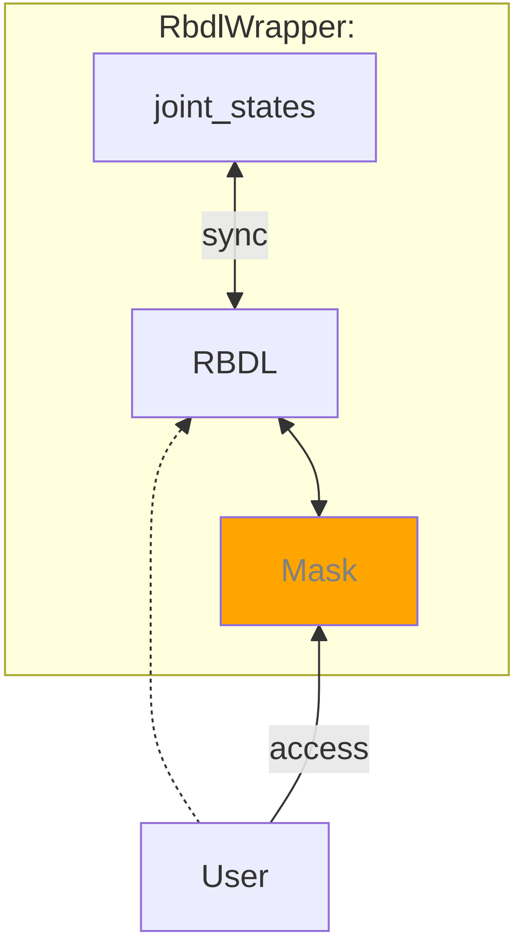
$$p_{\text{leftFoot}}^{\text{leftFoot}} \in \text{GND} \tag{12}$$

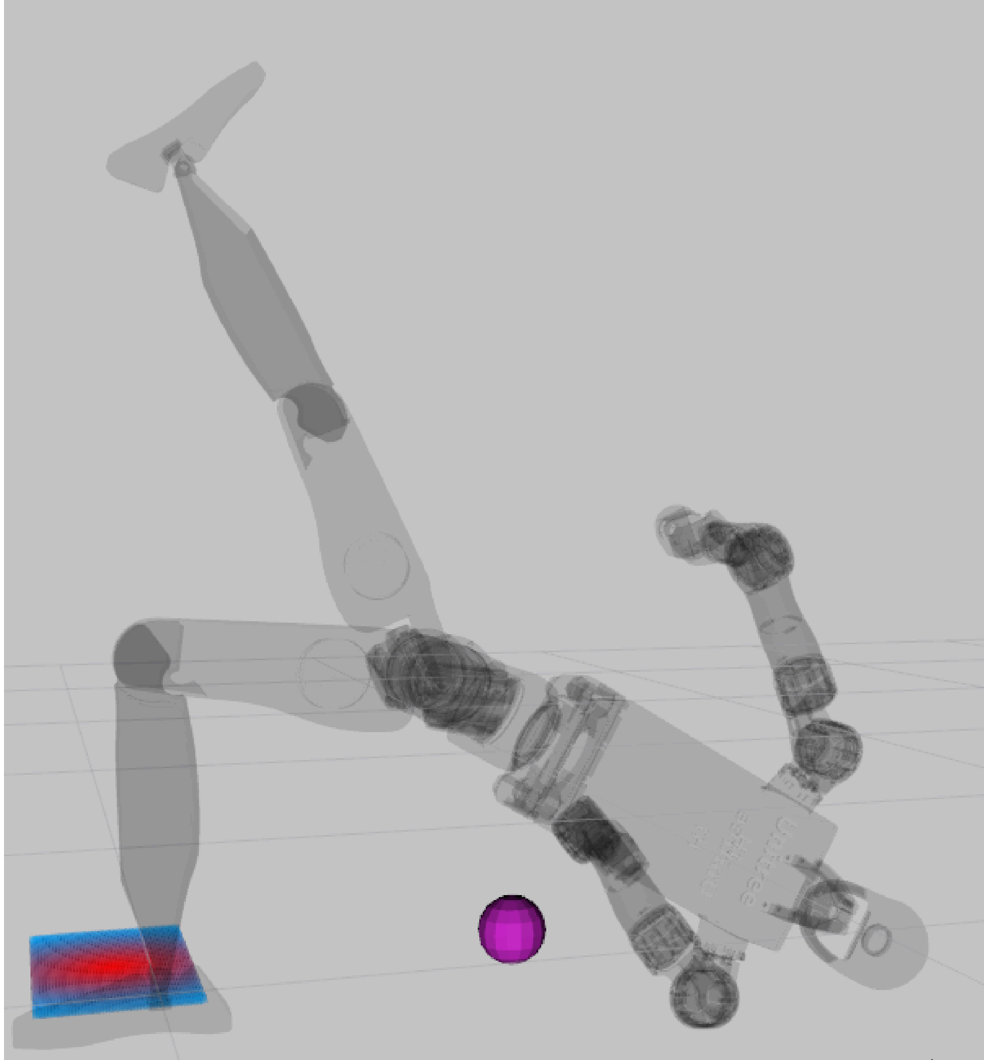
$$\text{AND} \quad p_{\text{rightFoot}}^{\text{leftFoot}} \in \text{GND} \tag{13}$$

$$p_{\text{CoP}}^{\text{leftFoot}} \in \text{PoS}^{\text{leftFoot}} \tag{14}$$

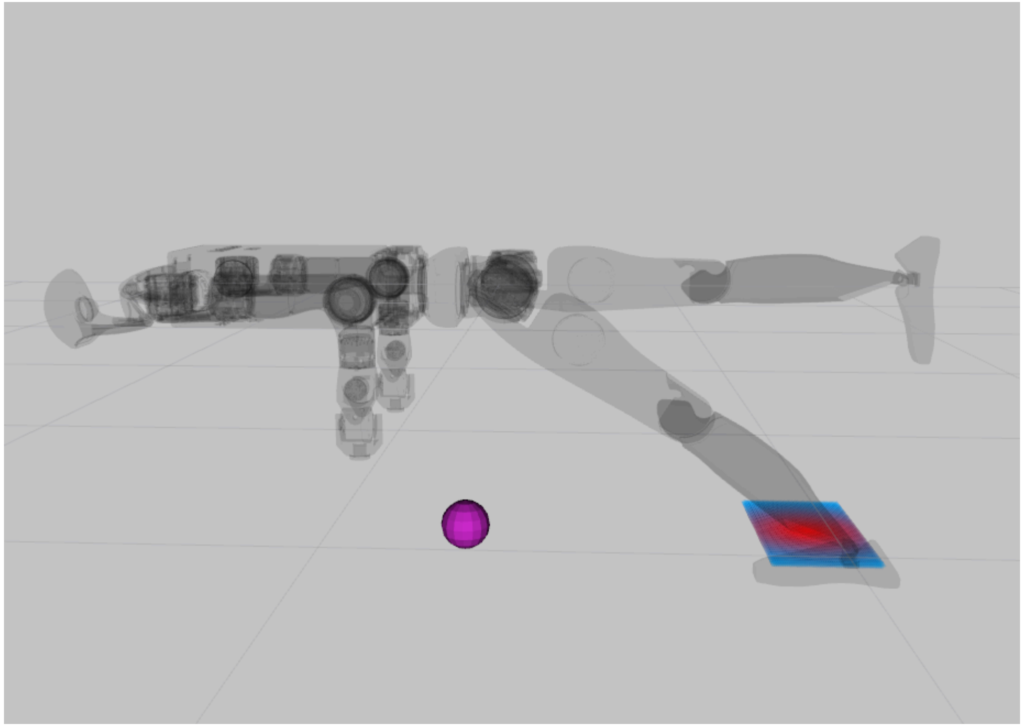
$$\xRightarrow{\text{static}} p_{\text{CoP}}^{\text{leftFoot}} := \begin{bmatrix} p_{\text{CoM}}^{\text{leftFoot}}.x \\ p_{\text{CoM}}^{\text{leftFoot}}.y \\ 0 \end{bmatrix} \tag{15}$$




RbdlWrapper:
joint_states
sync
RBDL
Mask
access
User



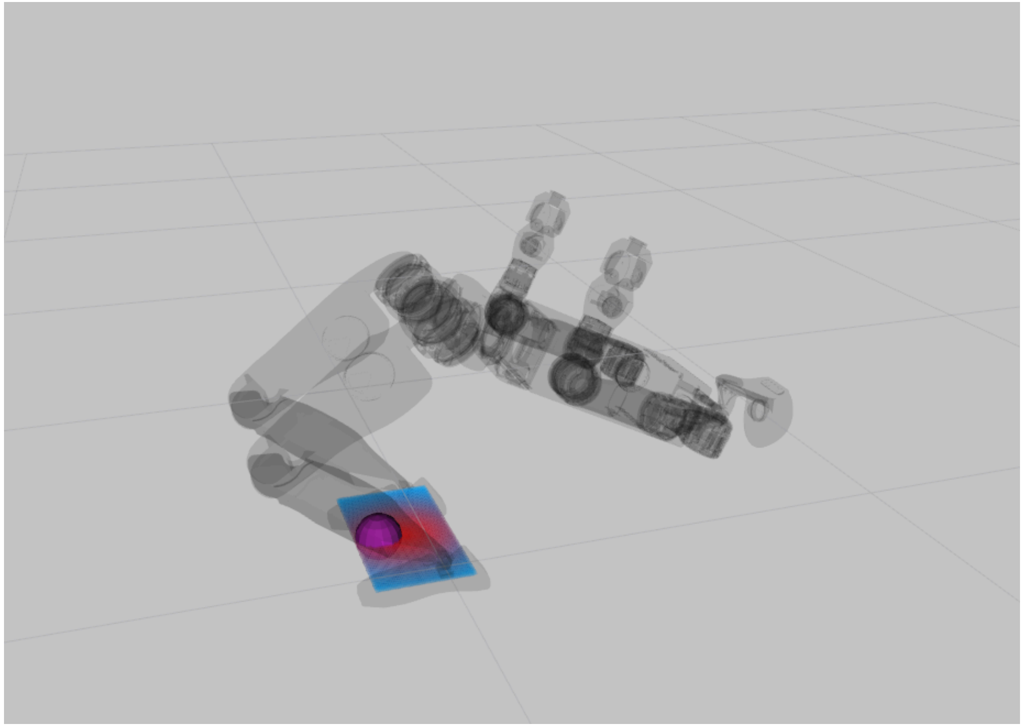
Just a Little Push 🥊



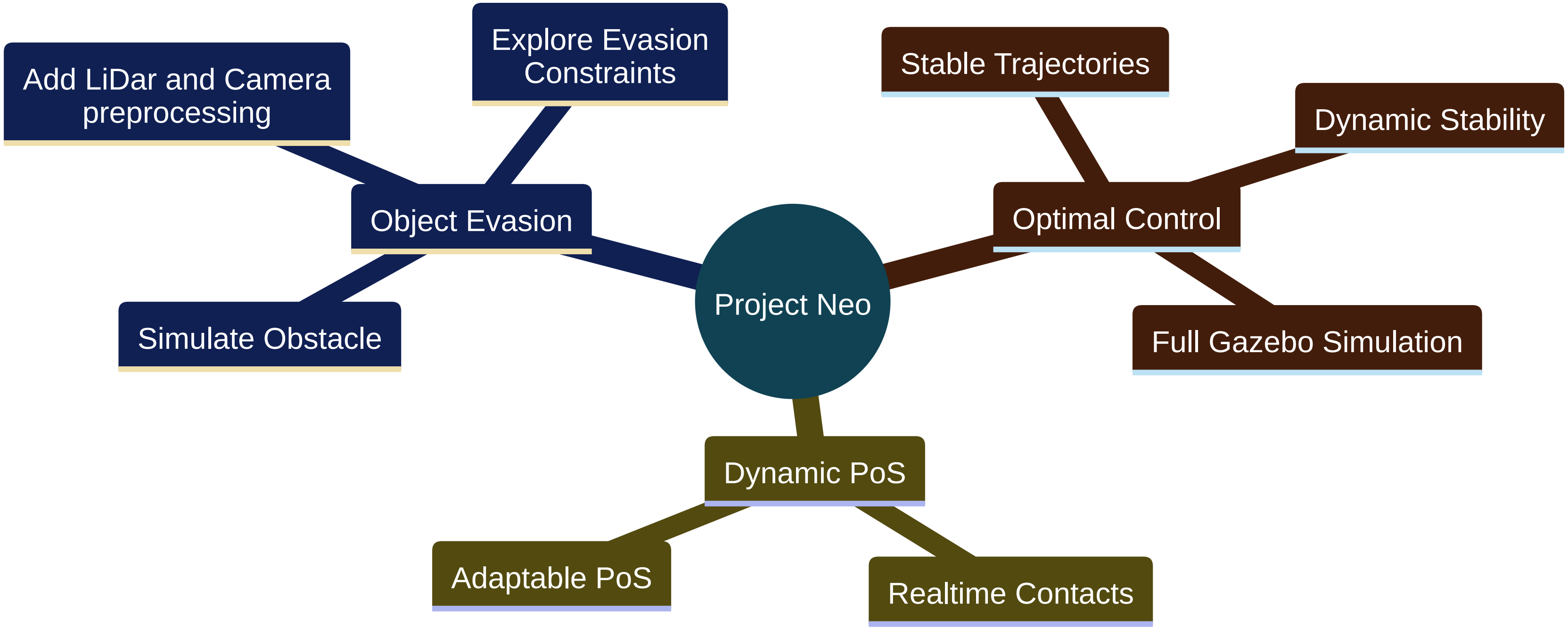
🔗 unsynced Legs + 🎱 bad Start Configuration

🥋 Live Demo 🥋

🎉Welcome `He1nz` the chosen one🎉



Future Work 🔭


Add LiDar and Camera preprocessing
Explore Evasion Constraints
Object Evasion
Simulate Obstacle
Project Neo
Stable Trajectories
Dynamic Stability
Optimal Control
Full Gazebo Simulation
Dynamic PoS
Adaptable PoS
Realtime Contacts

Thank you for following the White Rabbit.
🐇